

Automated Customer Reviews - NLP Project

Introduction

This project leverages NLP to automate classification, clustering, and summarization of Amazon product reviews.

Data Description

Dataset: Amazon Reviews

Preprocessing: cleaning, tokenization, rating mapping.

Review Classification

Model: BERT

Sentiment based on star ratings

Metrics: Accuracy, F1, Confusion Matrix.

Clustering

Sentence-BERT embeddings with KMeans (n=5)

Visualized using PCA.

Summarization

Models: T5/BART

Output: Top products, complaints.

Web Application

Built with Gradio

Includes classification, clustering, summarization modules

Deployed on AWS for accessibility.

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Challenges & Improvements

- Imbalanced dataset affecting model performance
- Long training times for large models
- Limited labeled data for fine-tuning
- Difficulty choosing optimal number of clusters
- Deployment configuration issues on AWS (ports, dependencies)

We improved the models iteratively and resolved deployment challenges.

Conclusion

Effective insights from automated reviews analysis. Models deployed and tested on AWS.

Appendix

GitHub Link, Code references.